

Maxwell's Equations in a Medium: Wave Equations, the Poynting Vector, and Poynting's Theorem

This companion to *Maxwell-wave-equations-poynting* (vacuum) treats a material medium. The structure is the same: write the source equations, take curls to get wave equations, and build the energy balance. The new ingredients are the macroscopic fields $\mathbf{D}$, $\mathbf{H}$, constitutive relations, free sources, and Ohmic loss.

1. Macroscopic Maxwell equations

In a material medium the microscopic fields are averaged over many atoms, introducing the **electric displacement** $\mathbf{D}$ and the **magnetic field intensity** $\mathbf{H}$. The source (macroscopic) equations are

$$\nabla \cdot \mathbf{D} = \rho_f$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

Here ρ_f and $\mathbf{J}_f$ are the **free** (unbound) charge and current. The bound charge and magnetization of the material are absorbed into $\mathbf{D}$ and $\mathbf{H}$. Note:

- Gauss's law for magnetism is unchanged: there are no magnetic monopoles in any medium.
 - Faraday's law is unchanged in form (the electric field is the same microscopic quantity).
 - The two "source" laws are rewritten with $\mathbf{D}$, $\mathbf{H}$ so that only free sources appear on the right.
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2. Constitutive relations

For a **linear, isotropic, homogeneous** medium the response is proportional:

$$\mathbf{D} = \varepsilon \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E}$$

$$\mathbf{B} = \mu \mathbf{H} = \mu_0 \mu_r \mathbf{H}$$

$$\mathbf{J}_f = \sigma \mathbf{E} \quad (\text{Ohmic conduction})$$

where ε_r is the relative permittivity, μ_r the relative permeability, and σ the electrical conductivity. The limit $\varepsilon \rightarrow \varepsilon_0$, $\mu \rightarrow \mu_0$, $\sigma \rightarrow 0$ recovers vacuum.

Two limiting cases are developed below:

- **Case A — lossless dielectric:** $\sigma = 0$, $\rho_f = \mathbf{J}_f = 0$.

- **Case B — lossy (conducting) medium:** $\sigma \neq 0$, $\rho_f = 0$ (free charge in a conductor decays to the surface, so the bulk is neutral).
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3. Case A — wave equation in a lossless dielectric

With $\sigma = 0$ and no free sources, the equations are

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{H} = \partial_t \mathbf{D}$$

with $\mathbf{D} = \varepsilon \mathbf{E}$, $\mathbf{B} = \mu \mathbf{H}$.

3.1 Wave equation for E

Step 1 — Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{H}) = -\frac{\partial}{\partial t}(\partial_t \mathbf{D}) = -\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Step 2 — Simplify the left-hand side with the curl-curl identity. The divergence term vanishes because $\nabla \cdot \mathbf{E} = \nabla \cdot (\mathbf{D}/\varepsilon) = (\nabla \cdot \mathbf{D})/\varepsilon = 0$:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$$

Step 3 — Equate:

$$\nabla^2 \mathbf{E} = \varepsilon \mu \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

3.2 Wave equation for H

Take the curl of the source-free Ampère law $\nabla \times \mathbf{H} = \varepsilon \partial_t \mathbf{E}$:

$$\nabla \times (\nabla \times \mathbf{H}) = \varepsilon \frac{\partial}{\partial t}(\nabla \times \mathbf{E}) = \varepsilon \frac{\partial}{\partial t}(-\partial_t \mathbf{B}) = -\varepsilon \mu \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

The left-hand side is $-\nabla^2 \mathbf{H}$ because $\nabla \cdot \mathbf{H} = (\nabla \cdot \mathbf{B})/\mu = 0$. Therefore

$$\nabla^2 \mathbf{H} = \frac{1}{v^2} \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

3.3 Plane-wave consequences

With $\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ the wave equation gives $\omega = v |\mathbf{k}|$ with the **phase velocity**

$$v = \frac{1}{\sqrt{\mu \varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$

and the **refractive index** $n = c/v = \sqrt{\mu_r \varepsilon_r}$ (for a non-magnetic medium $n \approx \sqrt{\varepsilon_r}$). Faraday's law gives

$$\mathbf{H} = \frac{\hat{\mathbf{k}} \times \mathbf{E}}{\eta}, \quad \mathbf{B} = \frac{\hat{\mathbf{k}} \times \mathbf{E}}{v}$$

so the waves are again **transverse** ($\mathbf{E} \perp \mathbf{B} \perp \hat{\mathbf{k}}$), with $|\mathbf{B}| = |\mathbf{E}|/v$ (generalizing the vacuum $|\mathbf{B}| = |\mathbf{E}|/c$). The ratio $|\mathbf{E}|/|\mathbf{H}|$ is the **intrinsic (characteristic) impedance**

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_0}{\sqrt{\mu_r \varepsilon_r}}, \quad \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 377 \, \Omega$$

4. Case B — wave equation in a lossy (conducting) medium

Now $\mathbf{J}_f = \sigma \mathbf{E}$ but $\rho_f = 0$. The only equation that changes is Ampère's law:

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t}$$

4.1 Damped wave equation

Take the curl of Faraday's law and substitute:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{H}) = -\frac{\partial}{\partial t}(\sigma \mathbf{E} + \varepsilon \partial_t \mathbf{E}) = -\sigma \mu \frac{\partial \mathbf{E}}{\partial t} - \varepsilon \mu \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

The left-hand side is still $-\nabla^2 \mathbf{E}$ (since $\nabla \cdot \mathbf{E} = 0$ in the neutral bulk). Hence

$$\nabla^2 \mathbf{E} = \varepsilon \mu \frac{\partial^2 \mathbf{E}}{\partial t^2} + \sigma \mu \frac{\partial \mathbf{E}}{\partial t}$$

This is the **damped (telegrapher's) wave equation**: the extra first-in-time term $\sigma \mu \partial_t \mathbf{E}$ damps the wave as it propagates. The same equation holds for $\mathbf{H}$.

4.2 Complex permittivity and complex wave number

For harmonic fields (time dependence $e^{-i\omega t}$, so $\partial_t \rightarrow -i\omega$) the equation becomes

$$\nabla^2 \mathbf{E} + \omega^2 \mu \left(\varepsilon + i \frac{\sigma}{\omega} \right) \mathbf{E} = 0$$

Define the **complex permittivity**

$$\varepsilon_c = \varepsilon + i \frac{\sigma}{\omega}$$

so that, for a plane wave $e^{i\mathbf{k} \cdot \mathbf{x}}$ (complex $\mathbf{k}$),

$$k^2 = \omega^2 \mu \varepsilon_c = \omega^2 \mu \varepsilon \left(1 + i \frac{\sigma}{\omega \varepsilon} \right)$$

Write $k = k_r + ik_i$ (propagation along $+x$): the real part k_r sets the phase and the imaginary part k_i sets the attenuation, $\mathbf{E} \propto e^{ik_r x} e^{-k_i x} e^{-i\omega t}$. Matching real and imaginary parts of k^2 :

$$k_r^2 - k_i^2 = \omega^2 \mu \varepsilon, \quad 2k_r k_i = \omega \mu \sigma$$

The dimensionless ratio $\sigma/(\omega \varepsilon)$ is the **loss tangent** $\tan \delta_{\text{loss}}$.

4.3 Good conductor ($\sigma \gg \omega \varepsilon$)

Here $\omega^2 \mu \varepsilon \ll \omega \mu \sigma$, so $k^2 \approx i \omega \mu \sigma$ and

$$k \approx \sqrt{\frac{\omega \mu \sigma}{2}} (1 + i), \quad k_r \approx k_i \approx \sqrt{\frac{\omega \mu \sigma}{2}}$$

The field decays as $e^{-x/\delta}$ with the **skin depth**

$$\delta = \frac{1}{k_i} = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{\rho}{\pi f \mu}} \quad (\rho = 1/\sigma)$$

The phase velocity is $v_p = \omega/k_r = \sqrt{2\omega/(\mu\sigma)}$, and the intrinsic impedance is $k_r \approx k_i$ giving $\eta \approx (1 + i)\sqrt{\omega\mu\sigma/2}$: the field penetrates only a few skin depths, and **H** lags **E** by 45° .

4.4 Lossy dielectric ($\sigma \ll \omega \varepsilon$)

Here $k_r \gg k_i$, so

$$k_r \approx \omega \sqrt{\mu \varepsilon}, \quad k_i \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\varepsilon}}$$

The wave travels almost at the lossless speed $v = 1/\sqrt{\mu \varepsilon}$ but is weakly attenuated; the attenuation per unit length is k_i (nepers/m).

5. Poynting's theorem and the Poynting vector in a medium

5.1 Derivation (general, valid for any medium + sources)

Use the two curl equations in macroscopic form:

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D}$$

Form $\mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H})$ and apply the identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H})$:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (-\partial_t \mathbf{B}) - \mathbf{E} \cdot (\mathbf{J}_f + \partial_t \mathbf{D}) = -\mathbf{H} \cdot \partial_t \mathbf{B} - \mathbf{E} \cdot \partial_t \mathbf{D} - \mathbf{E} \cdot \mathbf{J}_f$$

Rearranging,

$$\mathbf{E} \cdot \partial_t \mathbf{D} + \mathbf{H} \cdot \partial_t \mathbf{B} = -\nabla \cdot (\mathbf{E} \times \mathbf{H}) - \mathbf{E} \cdot \mathbf{J}_f$$

For a **linear, time-invariant, non-dispersive** medium ($\mathbf{D} = \varepsilon \mathbf{E}$, $\mathbf{B} = \mu \mathbf{H}$ with constant ε, μ), the left-hand side is a total time derivative:

$$\mathbf{E} \cdot \partial_t \mathbf{D} = \partial_t \left(\frac{1}{2} \varepsilon E^2 \right), \quad \mathbf{H} \cdot \partial_t \mathbf{B} = \partial_t \left(\frac{1}{2} \mu H^2 \right)$$

5.2 Energy density and Poynting vector

Define the **electromagnetic energy density** and the **Poynting vector** (the energy flux):

$$u = \frac{1}{2} \mathbf{E} \cdot \mathbf{D} + \frac{1}{2} \mathbf{B} \cdot \mathbf{H} = \frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2 = \frac{1}{2} \varepsilon E^2 + \frac{1}{2\mu} B^2$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\text{units: W/m}^2)$$

(These reduce to the vacuum expressions when $\varepsilon \rightarrow \varepsilon_0$, $\mu \rightarrow \mu_0$; note $\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$ in vacuum because $\mathbf{H} = \mathbf{B}/\mu_0$.)

5.3 Poynting's theorem

Substituting into the identity above gives the **differential form**

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_f$$

Integrating over a volume V bounded by a surface S with outward normal:

$$\frac{d}{dt} \int_V u dV = - \oint_S \mathbf{S} \cdot d\mathbf{a} - \int_V \mathbf{E} \cdot \mathbf{J}_f dV$$

Reading: the rate of change of stored electromagnetic energy in V equals (net power flowing *into* V through its surface) minus (power dissipated inside V). This is the local statement of energy conservation for fields in a medium.

5.4 Ohmic (Joule) loss

For an Ohmic medium $\mathbf{J}_f = \sigma \mathbf{E}$, so

$$\mathbf{E} \cdot \mathbf{J}_f = \sigma E^2 \geq 0$$

is the **Joule heating density** (power per unit volume converted to heat). Special cases:

- **Lossless, source-free dielectric** ($\mathbf{J}_f = 0$): $\partial_t u + \nabla \cdot \mathbf{S} = 0$ — exactly the vacuum continuity equation, with u and $\mathbf{S}$ in the medium form.
- **Conductor:** part of the flux is steadily converted to heat, which is why the wave attenuates over the skin depth.

6. Intensity of a plane wave in a dielectric

For a monochromatic plane wave traveling along $\hat{\mathbf{k}}$ in a lossless dielectric, $\mathbf{H} = (\hat{\mathbf{k}} \times \mathbf{E})/\eta$, so

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \frac{1}{\eta} \mathbf{E} \times \hat{\mathbf{k}}, \quad |\mathbf{S}| = \frac{E^2}{\eta}$$

Time-averaging over a period for $\mathbf{E} = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$ ($\langle \cos^2 \rangle = \frac{1}{2}$) gives the **intensity**

$$I = \langle S \rangle = \frac{E_0^2}{2\eta} = \frac{1}{2} \varepsilon v E_0^2$$

Compare with vacuum $I = E_0^2/(2\eta_0)$: a dielectric of index n carries more intensity for the same field amplitude, since $\eta = \eta_0/n$ (non-magnetic case). In a **lossy** medium the intensity decays as $I(x) = I_0 e^{-2k_i x}$ (the factor 2 because $I \propto E^2$).

7. Caveats for general media

The clean wave equation and the simple energy density above assume a **linear, isotropic, homogeneous, non-dispersive** medium. Outside that class the picture changes:

- **Dispersive** ($\varepsilon = \varepsilon(\omega)$, $\mu = \mu(\omega)$): $\mathbf{E} \cdot \partial_t \mathbf{D} \neq \partial_t (\frac{1}{2} \varepsilon E^2)$, so the stored energy is not simply $\frac{1}{2} \varepsilon E^2$. For a single harmonic, $u = \frac{1}{2} \frac{d(\omega \varepsilon)}{d\omega} E^2 + \frac{1}{2} \frac{d(\omega \mu)}{d\omega} H^2$ (the Brillouin formula). The wave equation still holds but $k = k(\omega)$, so different frequencies travel at different speeds (pulse dispersion).
- **Anisotropic** ($\mathbf{D} = \varepsilon \cdot \mathbf{E}$, ε a tensor): the $\nabla(\nabla \cdot \mathbf{E})$ term no longer decouples, the wave equation couples the field components, and the medium is birefringent (two polarization eigenmodes with different velocities).
- **Nonlinear** ($\mathbf{D} = \mathbf{D}(\mathbf{E})$): no linear wave equation; superposition fails, giving harmonic generation, self-focusing, etc.

8. Summary of results

Quantity	Lossless dielectric	Vacuum (limit)
Wave equation	$\nabla^2 \mathbf{F} = \mu \varepsilon \partial_t^2 \mathbf{F} = \frac{1}{v^2} \partial_t^2 \mathbf{F}$, $\mathbf{F} \in \{\mathbf{E}, \mathbf{H}\}$	<i>same</i> , $\mu = \mu_0$, $\varepsilon = \varepsilon_0$
Phase velocity	$v = 1/\sqrt{\mu \varepsilon} = c/\sqrt{\mu_r \varepsilon_r}$	$c = 1/\sqrt{\mu_0 \varepsilon_0}$
Refractive index	$n = c/v = \sqrt{\mu_r \varepsilon_r}$	1
Intrinsic impedance	$\eta = \sqrt{\mu/\varepsilon} = \eta_0/\sqrt{\mu_r \varepsilon_r}$	$\eta_0 \approx 377 \Omega$
Field ratio	$ \mathbf{B} = \mathbf{E} /v$, $ \mathbf{E} / \mathbf{H} = \eta$	$ \mathbf{B} = \mathbf{E} /c$
Damped wave eq. (lossy)	$\nabla^2 \mathbf{E} = \varepsilon \mu \partial_t^2 \mathbf{E} + \sigma \mu \partial_t \mathbf{E}$	—
Complex permittivity	$\varepsilon_c = \varepsilon + i\sigma/\omega$	—
Skin depth (good conductor)	$\delta = \sqrt{2/(\omega \mu \sigma)} = \sqrt{\rho/(\pi f \mu)}$	—
Energy density	$u = \frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2$	$u = \frac{1}{2} \varepsilon_0 E^2 + \frac{1}{2} \mu_0 H^2$
Poynting vector	$\mathbf{S} = \mathbf{E} \times \mathbf{H}$	$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$
Poynting's theorem	$\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_f$	$\partial_t u + \nabla \cdot \mathbf{S} = 0$
Momentum density	$\mathbf{g} = \mathbf{S}/c^2 = \varepsilon \mathbf{E} \times \mathbf{H}$ (<i>approx.</i>)	$\mathbf{g} = \varepsilon_0 \mathbf{E} \times \mathbf{B}$
Intensity (plane wave)	$I = E_0^2/(2\eta) = \frac{1}{2} \varepsilon v E_0^2$	$I = E_0^2/(2\eta_0)$